

cloud workloads, organisations can better prepare and implement effective protection strategies.

Key threats to cloud workloads

Cloud workloads face a myriad of threats that can compromise their security, integrity, and availability. Understanding these threats is crucial for implementing effective security measures.

Data breaches and ransomware:

These are critical threats where sensitive information is accessed or encrypted without permission, often leading to significant financial and reputational damage.

DDoS and phishing attacks:

These attacks disrupt service availability or trick users into revealing sensitive information, severely impacting cloud operations.

Cyberattacks: These include ransomware, supply chain attacks, insider threats, and cryptominers, each presenting unique challenges to cloud security.

Expansion of attack surfaces

The adoption of microservices and the increasing complexity of cloud architectures can significantly expand the attack surface, making cloud workloads vulnerable to unauthorised access and attacks.

Unmanaged attack surface:

Increased public exposure of microservices leads to higher risks of cyberattacks.

Misconfiguration: As cloud services evolve, the risk of misconfiguration grows, often due to varying default settings across multiple providers.

Human factor and insider threats

Human error and insider threats remain significant concerns, with potential for severe security breaches stemming from within the organisation.

Human error: Predicted to be the cause of 99% of cloud security failures.

Insider threats: These threats can arise from anyone within the organisation who has access to sensitive data and systems.

Advanced persistent threats and zero-day exploits

Sophisticated cyber threats such as APTs and zero-day exploits can bypass conventional security measures, requiring advanced detection and response strategies.

Advanced persistent threats (APTs):

These are complex, long-duration attacks aimed at stealing sensitive data.

Zero-day exploits:

These exploits take advantage of unpatched vulnerabilities, posing a significant risk even to well-secured cloud environments.

Challenges in identity and access management

Proper management of identities and access controls is foundational to securing cloud workloads, yet it remains a daunting task for many organisations.

Identity and access management (IAM):

Critical for defining and managing roles and permissions, yet complex to administer across large enterprises.

Unauthorised access and insecure APIs:

These can provide attackers easy paths to infiltrate cloud workloads if not properly secured.

Compliance and visibility issues:

Ensuring compliance with regulatory standards and maintaining visibility into cloud operations are pivotal for security but are often challenging due to the cloud's complex nature.

Cloud compliance: Organisations must continuously meet standards like PCI DSS and HIPAA, which protect sensitive data.

Lack of visibility: Traditional tools may not offer sufficient visibility in cloud environments, hindering effective monitoring and protection.

Misconfiguration and external data sharing risks

Misconfiguration of cloud settings is a leading cause of security breaches, and external data sharing introduces additional risks.

Misconfiguration: Commonly leads to data breaches, emphasising the need for robust cloud security posture management.

External sharing of data: While facilitating collaboration, it also poses risks if access controls are not properly managed.

Best practices for securing cloud workloads

Understanding your shared responsibility

Cloud workload security is a shared responsibility between the organisation, its customer, and the cloud provider. It's essential to understand that while the provider secures the infrastructure, the organisation is responsible for securing its data and its access. This model requires businesses to actively manage and secure several aspects of their cloud environment.

To safeguard their cloud workloads effectively, organisations must focus on the following areas.

Cloud management console and API keys: Ensure that all your cloud management tools and API SSH keys are secured against unauthorised access.

Virtual infrastructure and DevOps tools: Protect your virtual infrastructure and DevOps admin consoles and tools. This includes securing your DevOps pipeline code and SaaS applications admin accounts.

Cloud entitlements and permissions: Manage cloud entitlements carefully to prevent excessive permissions that could lead to security breaches.

Implementing multilayer security measures

Implementing multilayer security measures is crucial for robust cloud workload protection.